PROJECT BRIEF & HANDOVER · MOUSE-SCRATCH-RIG · V1.0 · 2026-07

Watching a cylinder full of mice, and catching every nose-scratch

A personal research rig that records mice from multiple synchronized cameras, tracks each animal, and automatically flags nose/face-scratching — throwing away the boring pixels so storage and analysis stay cheap. This document is the single source of truth: read it to understand the goal, the hardware, the software, the decisions, and the build plan. It is written to be picked up cold by a new collaborator — human or AI.

TYPE	REPO	PHASE NOW	NEXT
Personal project non-commercial	**erdanzhang/mouse-scratch-rig** private	**Hardware bring-up** 2-camera MVP	**M0 · M1** software, any OS

Goal · Hardware · Software · OSS stack · Worked example · Decisions · Milestones · Platform strategy · Risks

01 · OVERALL GOAL

What we are building, and what "done" means

Mice develop itch in many disease models, and **nose/face scratching** is a key readout — but scoring it by hand from hours of video is slow and subjective. The rig automates it: multiple cameras watch a transparent arena, software tracks each mouse with a stable ID, and a behavior model flags every scratching bout with a timestamp. A motion "key-pixel" gate keeps only the frames where something happens, so a mostly-sleeping cage doesn't cost terabytes or GPU-hours.

GOAL 1

Track every mouse

Multi-animal detection + stable IDs (mouse 1, 2, …) across occlusions and, later, across camera views in 3D.

GOAL 2

Flag nose-scratching

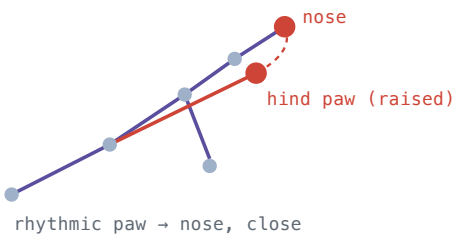
Automatically detect scratch bouts per mouse, with start/end timestamps and a confidence — output as an annotated video + JSON report.

GOAL 3

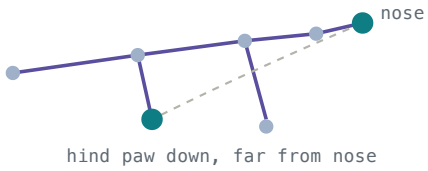
Stay cheap & self-improving

Drop idle pixels to shrink files; a human-in-the-loop labeling loop keeps raising accuracy from a small seed of labels.

scratching



not scratching · walking / front-paw grooming



the tracker outputs pose keypoints; scratching = a hind paw rises to the nose, rhythmically — the classifier watches that distance over ~45 frames.

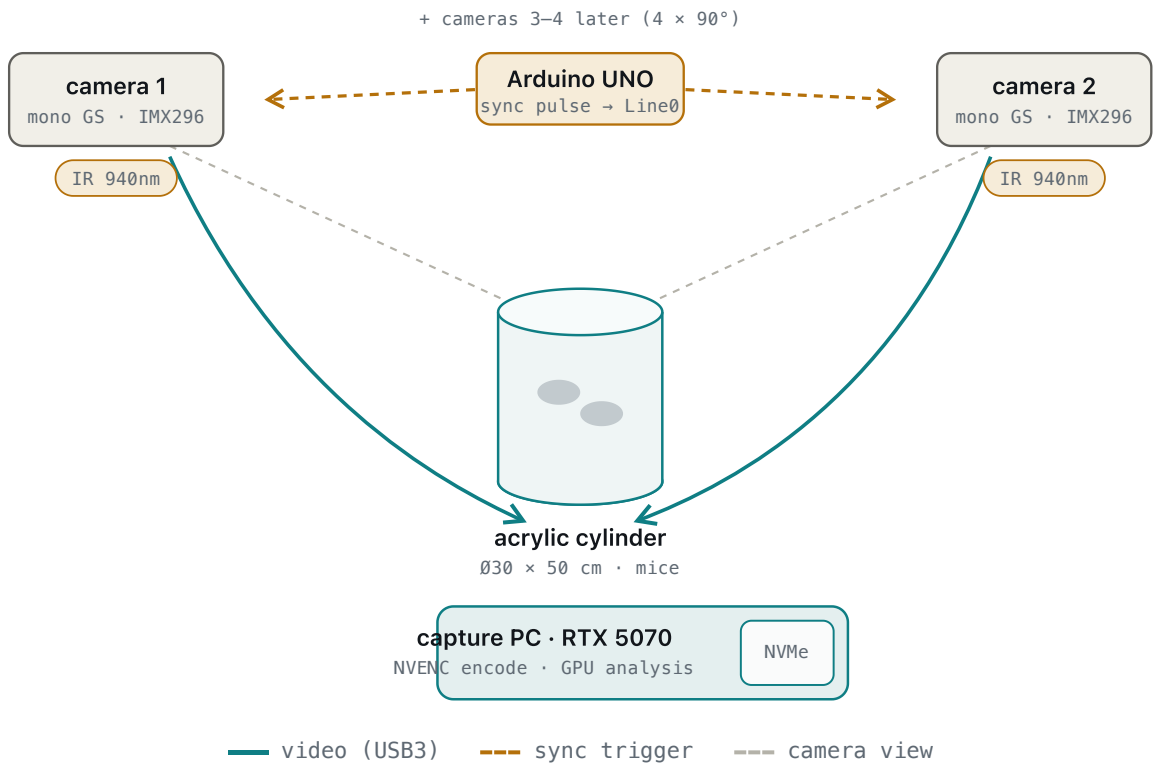
DEFINITION OF DONE (V1)

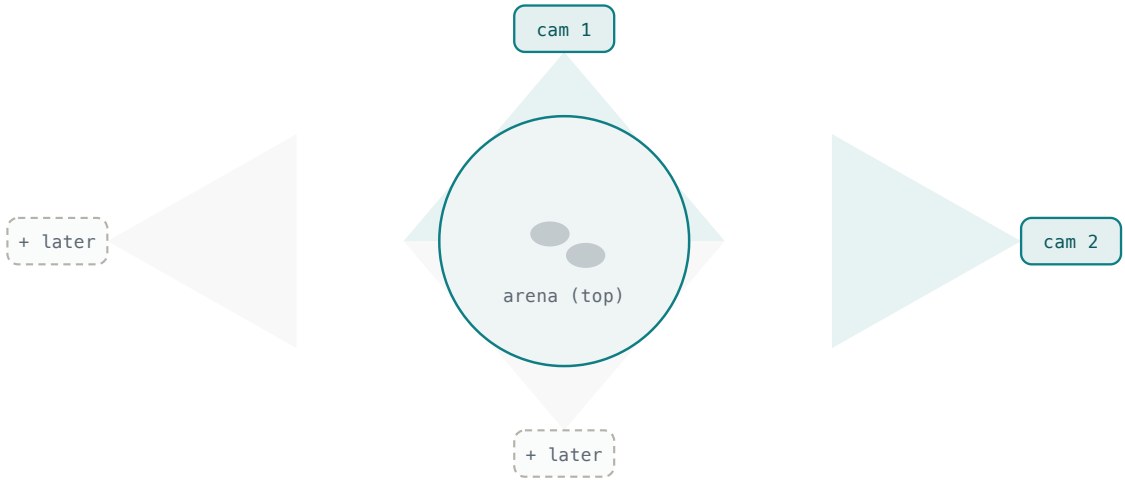
On a recorded session, the system outputs an annotated video (colored ID boxes; red boxes on confirmed scratching) plus a JSON report of scratch intervals per mouse — matching a human scorer within an agreed recall/precision target on a held-out validation set.

02 · HARDWARE ARCHITECTURE

How the physical rig fits together

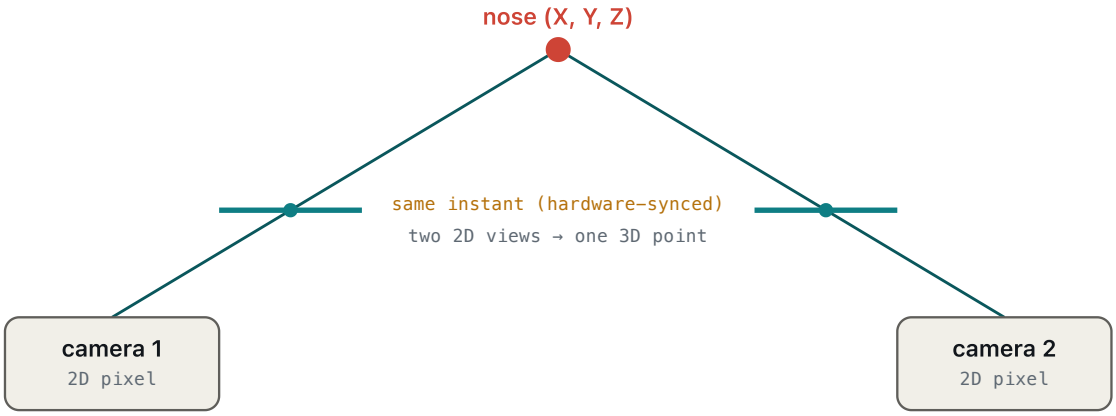
Cameras sit high around a clear cylinder, angled down to catch the face when a mouse rears to scratch. One Arduino pulse fires all shutters together so the views share a clock. Video travels over USB3 into a single PC whose GPU does the heavy lifting. Starts as 2 cameras; scales to 4.





top view — start with 2 cameras ~90° apart (a good triangulation baseline), then add 2 more for full surround coverage and occlusion robustness.

COMPONENT	CORE FUNCTION	CHOSEN PART	STATUS
Camera ×2→4	Blur-free frames of fast paw motion	Hikrobot MV-CS016-10UM (mono GS, IMX296)	arrived ×2
Lens	Fits the Ø30 cm arena in frame	C-mount 6–8 mm, ≥2 MP	arrived ×2
Sync	Fires all shutters on one clock	Arduino UNO + 6-pin IO cable (MV-ACC-01-2101)	arrived
Data link	Streams video to the PC	USB3 3 m, screw-lock; separate controllers	arrived ×2
IR light	Lights the dark phase, invisible to mice	940 nm illuminator + diffuser	pending
Mounts + arena	Rigid, fixed geometry for calibration	Arms/rail + acrylic cylinder Ø30×50	pending
Compute	Encode, track, gate, classify	PC + RTX 5070 (Win/Ubuntu) + NVMe ≥1 TB	the lab PC
Animals	The subject	Lab mice (white)	later

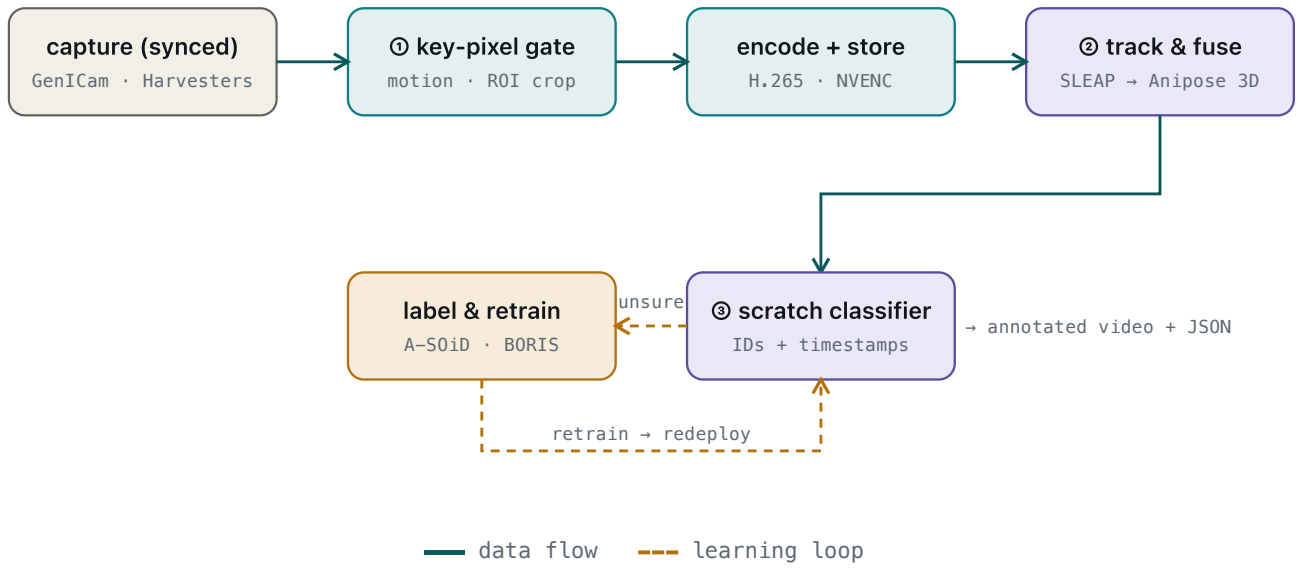


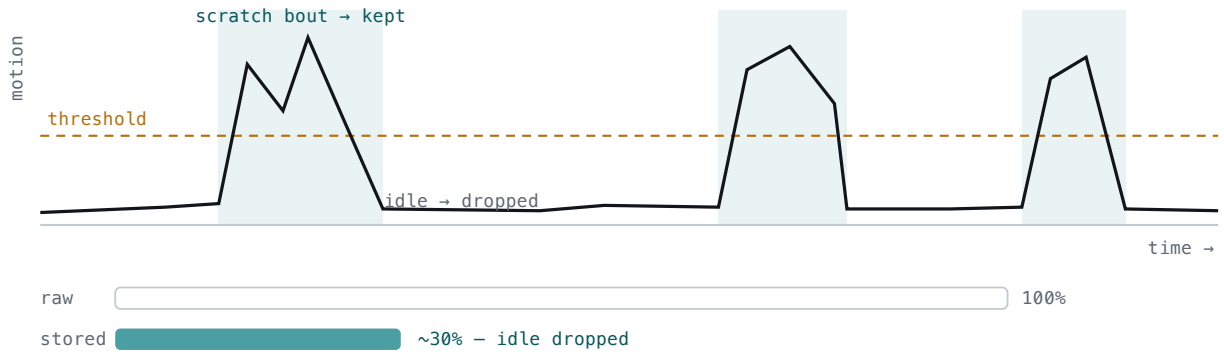
calibration measures where each camera sits; then every keypoint seen in both synced views is triangulated into a single 3D position. This is what "fusing the cameras" means.

03 · SOFTWARE ARCHITECTURE

From four lenses to one improving model

One continuous loop: capture in lock-step, discard pixels that don't matter on the GPU, analyze only what's left, and route the model's own uncertainty back to a human for labeling.





the motion gate keeps only frames above threshold; a mostly-sleeping cage collapses to a fraction of the raw size — less storage, less GPU to analyze.

MODEL ①

Track & fuse

Per-camera pose/detection, then triangulate the synced views into one 3D skeleton per mouse.

MODEL ②

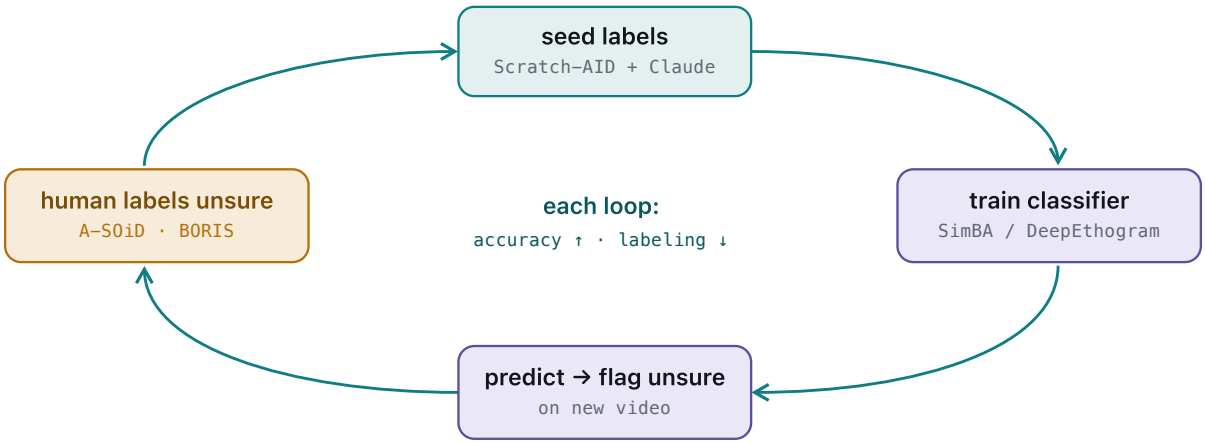
Key-pixel gate

Motion energy per mouse-ROI marks sleeping frames non-key → drop fps/bitrate; analyze only active clips.

MODEL ③

Label & self-improve

Seed labels → classifier flags low-confidence clips → human labels those → retrain. Accuracy climbs, effort falls.



start from a few seed labels; the model bulk-labels and only asks a human about clips it's unsure of — quality rises while human effort falls each round.

04 · OPEN-SOURCE STACK

Every model is open — and free for this project

This is a personal, non-commercial project, so the entire stack runs at zero licence cost (the copyleft/AGPL/academic terms only bite inside a commercial product). Preference order still favors permissive licences where there's a choice.

TOOL	ROLE	LICENCE
Harvesters + GenICam	Vendor-neutral camera capture (USB3 Vision)	Apache-2.0
Hikvision MVS / SDK	Camera driver + GenTL producer, trigger config	Vendor (free)
OpenCV	Motion gate, ChArUco calibration, triangulation	Apache-2.0
FFmpeg	H.265 encode (NVENC / VideoToolbox / x265)	LGPL/GPL
SLEAP	Multi-animal pose (2D per view)	BSD-3
DeepLabCut	Alt pose estimation	LGPL-3.0
Anipose	Multi-view → 3D triangulation	BSD-2
ByteTrack	Multi-object tracking / stable IDs	MIT
A-SOiD	Active-learning loop	Clear BSD
SimBA / BORIS / DeepEthogram	Behavior labeling + classifiers	GPL / academic
Scratch-AID	Reference scratch CRNN (bootstrap)	Open (academic)
Claude API	VLM auto-labeler to seed the loop	Paid API

05 · WORKED EXAMPLE

Follow one scratch through the system

#	STAGE	WHAT HAPPENS
1	**Trigger fires**	The Arduino sends one pulse; all cameras' global shutters expose the same instant — several views of a mouse mid-scratch.
2	**Into the PC**	USB3 delivers the frames to one grabber, tagged with a shared frame index.
3	**Key-pixel gate**	Motion energy in the mouse's box is high → the frame is **key**. A sleeping mouse elsewhere is cropped out / dropped.
4	**Encode**	NVENC writes only the active crops to H.265 — small file, full quality where it matters.
5	**Track & fuse**	Each view is pose-tracked; synced frames fuse to one 3D skeleton (or the clearest single view is picked).
6	**Classify**	Hind-paw-to-nose geometry over ~45 frames scores as **scratching** → logged with mouse ID + timestamp.
7	**Learn**	If the model was unsure, the clip enters the A-SOiD queue; a human confirms, the model retrains, next time it's certain.

06 · DECISIONS & RATIONALE

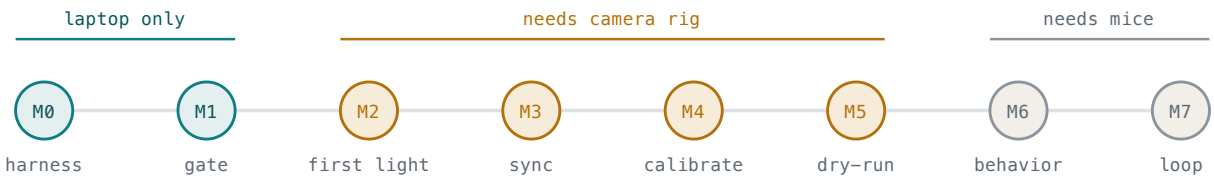
Choices already made (and why)

DECISION	CHOICE	WHY
Detector (no-training start)	Classical blob + Claude VLM; YOLO/SLEAP later	YOLO-World failed on the screen-recorded test footage; blob + VLM needs no labels to start.
Camera shutter	Global shutter	Scratching is fast; rolling shutter smears the paw.
Mono vs color	Mono for IR; color OK only under visible light	Under 940 nm IR a color sensor yields no real colour and is dimmer/softer.
IR wavelength	940 nm (not 850 nm)	940 nm is invisible to mice and doesn't disturb circadian rhythm; 850 nm glows red.
Interface	USB3 (GigE if PC is far)	Best bandwidth for a compact rig, straight into one PC.
Scale	Start 2 cameras → grow to 4	All software is camera-count-agnostic; adding cameras just re-runs calibration.
3D vs 2D	2D best-view first; upgrade to Anipose 3D	Scratch-AID hit ~97% from one view; 3D is a superset — keep frames synced to enable it later.
Compute location	RTX 5070 PC (Win/Ubuntu), not the Mac	Camera SDK + NVENC need it; Mac is for cross-platform development.

07 · IMPLEMENTATION PLAN

Milestones — each demo-able and end-to-end testable

Ordered by dependency. **M0–M1 need nothing but a laptop**; M2–M5 need the cameras/rig; M6–M7 need mice. Every milestone ships something you can run and a test that proves it.



dependency order — M0–M1 run today on any laptop; M2–M5 unlock with the rig; M6–M7 with the animals. Every node ships a runnable demo and an e2e test.

M0 Repo & platform harness

DO NOW · ANY OS

DELIVERABLE

Scaffolded repo; `mouse-rig report` prints platform capabilities (OS, encoders, torch device, camera producer).

DEMO

Run `report` on Mac → shows VideoToolbox/MPS; on the lab PC → shows NVENC/CUDA.

E2E TEST

`pytest` capabilities smoke test green on macOS and Windows.

M1 Key-pixel filter (file source)

DO NOW · ANY OS

DELIVERABLE

`mouse-rig gate <video>` → motion-gated, ROI-cropped, HW-encoded output + JSON stats.

DEMO

Run on the existing sample clip; report the storage reduction %.

E2E TEST

Assert idle frames dropped, output smaller than raw, and it plays back.

M2 Camera first light + Python grab

NEEDS CAMERAS

DELIVERABLE

GenICam `FrameSource`; `capture --source genicam:SERIAL` grabs frames.

DEMO

Live single-camera preview; saved sharp frames.

E2E TEST

Grab N frames headless; assert count + sharpness (variance-of-Laplacian).

M3 Dual capture + Arduino sync

NEEDS CAMERAS

DELIVERABLE

Dual capture with shared frame index; Arduino sketch; trigger config.

DEMO

LED/timer sync-test video showing aligned frames.

E2E TEST

Automated check: same-index frames match on a known signal within tolerance.

M4 Calibration + triangulation (fuse) PRACTICE NOW · REAL ON MOUNTS

DELIVERABLE	DEMO	E2E TEST
`calibrate` (ChArUco) → intrinsics/extrinsics; `triangulate` a marker to 3D.	Move a marker 10 cm → reconstructed 3D moves ~10 cm.	Reprojection error < 1 px; triangulation error under threshold on known geometry.

M5 Integrated capture dry-run NEEDS RIG

DELIVERABLE	DEMO	E2E TEST
Config-driven pipeline: synced capture → gate → encode → store + live preview.	1-hour dry-run pointed at a moving toy.	1 h run: assert zero dropped frames and storage rate within bound.

M6 Behavior model bootstrap NEEDS MICE

DELIVERABLE	DEMO	E2E TEST
Pose/tracking on real mice; scratch classifier v0 seeded by Scratch-AID + Claude auto-labels.	Annotated video: IDs + red SCRATCHING boxes + JSON report (the POC, on real footage).	Recall/precision vs a hand-labeled validation set (target ≥ 90% recall).

M7 Active-learning loop NEEDS MICE + M6

DELIVERABLE	DEMO	E2E TEST
A-SOiD uncertainty queue → human label → retrain → redeploy, scripted.	Accuracy climbs across 2–3 iterations with fewer new labels each round.	Measured accuracy gain + labeling-effort drop across iterations.

08 · PLATFORM-NEUTRALITY STRATEGY

One codebase, Mac or Windows

The pipeline is pure Python/OpenCV and runs anywhere. Three things are platform-specific and are hidden behind interfaces, feature-detected at runtime, and documented in `docs/PLATFORM.md`. Develop on the Mac with file/webcam sources; run the real cameras + NVENC on the lab PC — same code.

CONCERN	INTERFACE	MACOS BACKEND	WINDOWS / LINUX + RTX 5070
Frame source	`FrameSource`	File / Webcam (dev)	GenICam (Harvesters) — real cameras
Video encode	`Encoder`	VideoToolbox / x265	NVENC (hevc_nvenc) / x265
Model compute	`get_device()`	MPS / CPU	CUDA

THE RULE

No `import torch` or camera SDK at module top level — everything platform-specific is a guarded, lazily-imported backend chosen by a runtime capability probe. Base install (`numpy`, `opencv-python`, `pyyaml`) runs on any OS; capture and model dependencies are optional extras. `mouse-rig report` tells you what a given machine supports.

09 · CRITICAL INFORMATION & RISKS

Things a new collaborator must know

- **Secrets never in git.** API keys live in `.cred` / env vars, git-ignored. The repo is fresh and clean of the earlier POC's credentials.
- **RTX 5070 has a single NVENC encoder.** Fine for 2–4 × 1080p streams; a 5070 Ti/5080 (dual encoder) is the upgrade if high-fps 3D + live encode ever run together.
- **Calibration is only valid while cameras are fixed.** Do the real calibration after rigid mounting; a bump means recalibrate.
- **IR shifts focus.** Refocus when the 940 nm light replaces room light.
- **Animal welfare.** Clear the setup (esp. continuous IR) with local animal-care / veterinary / IACUC before any mice; IR images through the dark phase — it does not replace the normal light/dark cycle.
- **No mice yet.** M0–M5 use stand-ins (file/webcam, a moving toy, a ChArUco board); only M6–M7 need animals.

COMPANION DOCUMENTS

Detailed hardware BOM & sourcing → the hardware proposal. Step-by-step bring-up checklist → the bring-up runbook. This brief is the top-level context that ties them together.

mouse-scratch-rig · project brief v1.0 · personal / non-commercial · maintained as the single source of truth. References: Anipose (Cell Reports 2021), SLEAP (Nature Methods 2022), A-SOiD (Nature Methods 2024), Scratch-AID (eLife 2023), DeepEthogram (eLife 2021).